

Brijesh Kumar

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[IIT Bombay, Mumbai, India](#)

SUMMARY

- 3⁺ years of experience in clean-room processing at [IIT Bombay Nano-fabrication facility](#) and [FOTON lab, TIFR](#).
- 2⁺ years of experience in handling and operating equipments at [SAIF/CRNTS, IIT Bombay](#).
- Thorough understanding of the photonics domain and its ecosystem.
- Experienced in multi-team projects and coordinating cross-team activities.
- Established the [research-grade LOQM-Lab \(PI Prof. Anshuman Kumar\)](#) in rigorous scientific exploration.

PROFESSIONAL EXPERIENCE

- [Indian Institute of Technology Bombay \(IITB\)](#) Mumbai, India
Research Associate, LOQM - lab, Prof. Anshuman Kumar January 2026 - Present
- [Research School of Physics, The Australian National University \(ANU\)](#) Canberra, Australia
Visiting Research Student – Material Physics, under Prof. Yuri Kivshar Feb 2025 - Jan 2026

AREAS OF INTEREST

My research advances quantum photonic technologies and quantum sensing by engineering light–matter interactions in resonant nanophotonic systems. I design chiral metasurface architectures supporting high-Q Bound States in the Continuum (BICs), integrated with van der Waals materials (hBN, TMDCs), to enable chiroptical sensing and on-chip quantum photonic devices.

Bound States in the Continuum (BICs), Metasurfaces, Chiral Photonics, Van der Waals Materials (hBN, TMDCs), Quantum Photonics, Quantum Sensing

EDUCATION

- [Indian Institute of Technology Bombay \(IITB\)](#) Mumbai, India
PhD, LOQM - lab, Prof. Anshuman Kumar January 2021 - April 2026
Thesis title “Bound State in Continuum Metasurfaces: Fundamentals, Chirality, and Sensing”
- [Indian Institute of Technology Kharagpur \(IITKgp\)](#) Kharagpur, India
Master of Technology (M.Tech); CGPA: 8.79/10 July 2018 - June 2020
Master Thesis “Characterization of Dye-Doped Poly(methyl methacrylate) Thin Films and Fabrication of Microdisc Resonators by Solvent Immersion Imprint Lithography”
- [Bareilly College, Rohilkhand University](#) Bareilly, India
Master in Science (MSc.) in Physics; 68.2% July 2015 - June 2017
- [Bareilly College, Rohilkhand University](#) Bareilly, India
Bachelor in Science (BSc.) in Physics; 68.6% July 2012 - June 2015

JOURNAL PUBLICATIONS

1. Mangababu Akkanaboina*, Rohit Kumar*, **Brijesh Kumar*** et al, “Deterministic Single-Photon Emitter Arrays in Hexagonal Boron Nitride by Carbon-Assisted Focused Ion Beam Engineering” [arXiv preprint, 2026](#)
2. **Brijesh Kumar** et al “Intrinsic chirality of dielectric metasurfaces unlocked by resonant chiral modes” [Nano Lett. 2026;](#)
3. Ivan Toftul*, **Brijesh Kumar*** et al “Optical chirality of membrane metasurfaces with broken in-plane symmetry” [arXiv preprint, 2026](#) **Under review in ACS Photonics**
4. **Brijesh Kumar** et al “Maximal optical chirality via mode coupling in bilayer metasurfaces”, [ACS Photonics 2025 12 \(12\), 6717-6727](#)
5. **Brijesh Kumar** et al, “A universal and stable metasurface for photonic quasi bound state in continuum coupled with two dimensional semiconductors”, [Journal of Physics D: Applied Physics, Volume 56, Number 42 \(2023\)](#).
6. **Brijesh Kumar** et al, “High-quality-factor dye-doped polymeric microdiscs fabricated by soft imprint lithography.” [The European Physical Journal Special Topics 231 \(4\), 781-789, 2022](#)

7. **Brijesh Kumar**, et, al “Sensing with Broken Symmetry: Revisiting Bound States in the Continuum”, [arXiv preprint arXiv:2507.05676](#), 2025
8. Furkan Kuruoglu, Samir Rosas, Yihong Chen, **Brijesh Kumar** et al, “Dynamically tunable membrane metasurfaces for infrared spectroscopy and strong light-matter interactions”, [Light: Science & Applications](#) **15** (1), 269
9. Parul Sharma, **Brijesh Kumar**, et. al., “Exceptional point sensing via energy loss profile in a non-Hermitian system”, [Sensors and Actuators A Physical](#), 2024, 115681
10. Nihar Ranjan Sahoo, **Brijesh Kumar**, et. al., “Polaritons in photonic hypercrystals of van der Waals materials”, [Adv. Funct. Mater.](#) 2024, **34**, 2316863
11. Lekshmi Eswaramoorthy, Parul Sharma, **Brijesh Kumar**, et, al “Focused ion beam polishing based optimization of high-Q silica microdisk resonators”, [Phys. Scr.](#) **101** 045513, 2026
12. Lekshmi Eswaramoorthy, Parul Sharma, **Brijesh Kumar**, et, al “Protected Ion Beam Fabrication of Two-Dimensional Transition Metal Dichalcogenides based Photonic Devices”, [preprint arXiv:2510.26637](#)
13. Kishor Kumar Mandal, Anuj Kumar Singh, **Brijesh Kumar**, et. al., “Emission engineering in monolithically integrated silicon nitride microring resonators”, [ACS Materials Letters](#), volume **6** Pages 1831-1840, 2024
14. Nihar Ranjan Sahoo, SS Prasath, **Brijesh Kumar** et al, “Twisted hyperbolic van der Waals crystals for full Stokes mid-infrared polarization detection”, [J. Phys. D: Appl. Phys.](#) **57** 505104, 2024.
15. Kishor Kumar Mandal, Yashika Gupta, **Brijesh Kumar**, et al, “A photonic integrated chip platform for interlayer exciton valley routing”, [Journal of Applied Physics](#) **133**, 123104 (2023)
16. Anuj Kumar Singh, Utkarsh, Pablo Tieben, Kishor Kumar Mandal, **Brijesh Kumar** et al, “Plasmonic-Strain Engineering of Quantum Emitters in Hexagonal Boron Nitride”, [Adv. Mater. Interfaces](#) **2025**, **12**, 2500071.
17. Anuj Kumar Singh, Kishor K. Mandal, Yashika Gupta, Abhay Anand VS, Lekshmi Eswaramoorthy, **Brijesh Kumar** et al, “Low-Cost Plasmonic Platform for Photon-Emission Engineering of Two-Dimensional Semiconductors”, [Physical Review Applied](#) **19**, 044012 (2023).

PEER REVIEW EXPERIENCE

- **Reviewer, Optics Letters**, (8 reviews)
- **Reviewer, Photonics Research**, (4 reviews)
- **Reviewer, Optical Materials Express**, (1 reviews)
- **Reviewer, Advanced Science, Wiley**, (1 reviews)

INVITED TALKS AND POSTER PRESENTATION

1. **Contributed work poster on**, “Emission engineering via the silicon nitride photonic platform” [Integrated Optics: Devices, Materials, and Technologies XXVIII](#), Pages PC1288917, 2024
2. **Contributed work poster on**, “Engineering of excitonic emission and valley polarization enhancement in two-dimensional semiconductors” [2D Photonic Materials and Devices VII](#), SPIE, 12888, 46-50, 2023
3. **Presented poster on**, “Integration of 2D exciton with photonic bound state in continuum.”, [SPIE Optics + Optoelectronics Conference](#), Prague, Czech Republic, April 2023.
4. **Presented poster on**, “Exceptional point-based sensing via nanostructured photonic waveguide.”, [SPIE Optics + Optoelectronics Conference](#), Prague, Czech Republic, April 2023.
5. **Contributed work poster on**, “Valley selective routing of interlayer excitons in two-dimensional semiconductor heterostructures via microresonators.” [SPIE Optics + Optoelectronics Conference](#), Prague, Czech Republic, April 2023.
6. **Contributed work presented as poster on**, “Engineering photoluminescence using low-cost platform in two-dimensional transition-metal dichalcogenides.” [SPIE Optics + Optoelectronics Conference](#), Prague, Czech Republic, April 2023.
7. **Contributed work presented as poster on**, “Selective Coupling of Dark and Bright Excitons in 2D Transition Metal Dichalcogenide Alloys to Planar Microstructures” [Frontiers in Optics + Laser Science](#) 2021
8. **Presented poster on**, “Dye-Doped Poly(Methyl Methacrylate) Microdisc Resonators Fabricated by Solvent Immersion Imprint Lithography”, IONS 2020, India

POSITIONS OF RESPONSIBILITY AND LEADERSHIP

- **Managing Photonics Student Chapter IIT Bombay** Mumbai, India
Optica and IEEE professional society for optics and photonics technology Sept. 2021 - 2023
- **Teaching Assistant, During PhD tenure** Mumbai, India
- *Course Instructor, Department of Physics, IIT Bombay* Jan. 2020 - 2022
- **Teaching Assistant** NPTEL course
- *Course TA for Live interaction with the student* July 2022 - Dec 2022 and July 2023 - Sep 2023
- **System operator of Raith 150** Mumbai, India
- *Tool management, IIT Bombay Nano-fabrication Facility (IITBNF)* July 2023 - July 2025
- **Organizing committee member:** LASER World of PHOTONICS INDIA
- *In Organisation committee of PLAIM conference* Bombay Exhibition Centre, Mumbai, India, December 2022.

HONORS AND AWARDS

- **Prime Minister research fellowship**, conducted by Ministry of Education Government of India, January 2022.
- **Institute travel grant (IIT Bombay)** to attend an international conference (SPIE Optics + Optoelectronics, Prague, Czech Republic), 2023.
- Selected to attend **Siegman International School on Lasers** conducted by Optica foundation (awarded travel grant), hosted in Poland, Europe, 2023.
- **Junior Research Fellowship**, Indian Institute of technology IIT Bombay.
- **GATE Master's fellowship**, conducted by Indian Institutes of Technology Kharagpur, India, qualified in 2018.

TECHNICAL SKILLS

- **Semiconductor processing:** Optical lithography, electron beam lithography (RAITH-EBL), focused ion beam-scanning electron microscopy (FIB-SEM), sputtering, inductively coupled plasma reactive ion etching (ICP-RIE), Deep-RIE (DRIE), plasma enhanced chemical vapour deposition (PE-CVD), wet etching.
- **Physical characterization:** Scanning electron microscope (SEM), optical microscope, profilometer, ellipsometry, UV-Vis absorption, X-ray photoelectron spectroscopy (XPS), and atomic force microscope (AFM).
- **2D material processing:** Mechanical exfoliation of 2D van der Waals (vdW) materials using scotch tape and polydimethylsiloxane (PDMS)-assisted dry transfer.
- **Simulation and programming:** Ansys-Lumerical (FDTD), COMSOL Multiphysics, Python, MATLAB.
- **Layout editor:** CleWin, KLayout, Raith GDSII editors.
- **Graphics and data analysis:** Blender, VESTA, OriginLab, ImageJ.
- **Soft skills:** Leadership, journal writing (LaTeX, Microsoft Office), technical presentation, interpersonal communication.

CONFERENCES AND WORKSHOPS

- **SemiconIndia 2023, Catalyzing India's Semiconductor Ecosystem**, addressed by Hon'ble Prime Minister Shri Narendra Modi, Gujarat, India, July 2023.
- **SPIE Optics + Optoelectronics**, Prague, Czech Republic, April 2023.

REFEREES

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